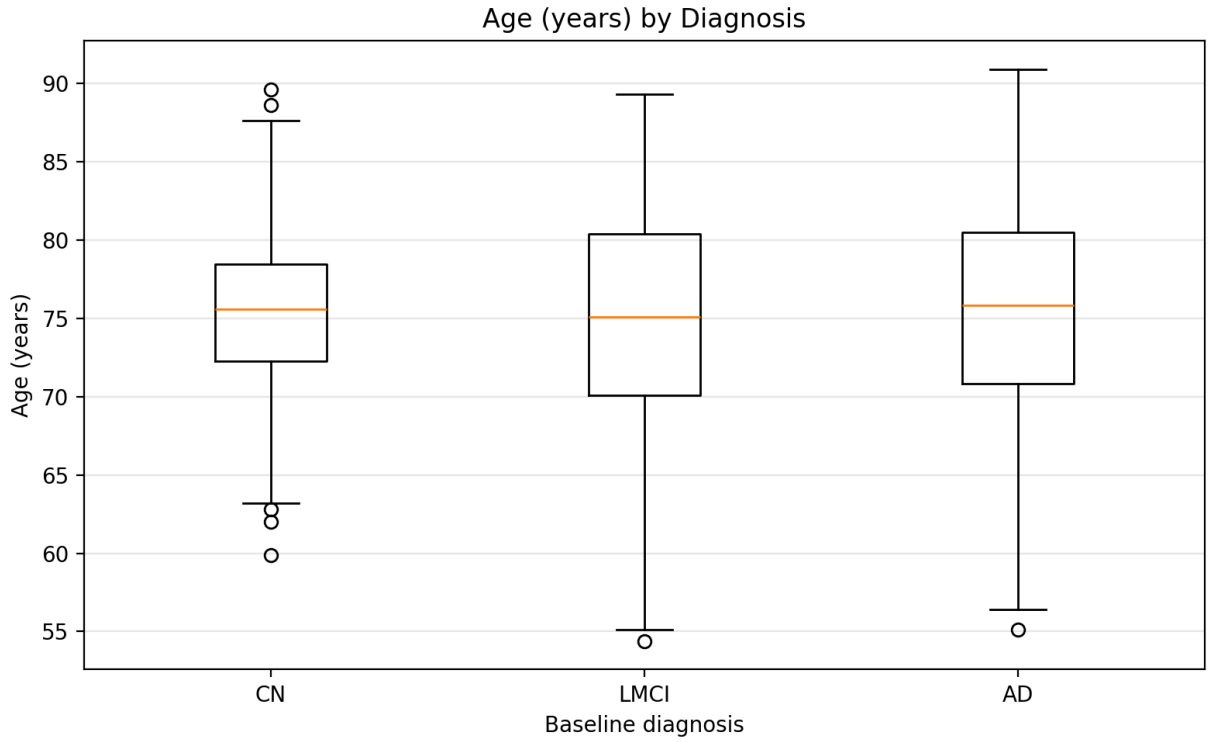


Statistic	**CN**	**LMCI**	**AD**
Sample Size	228	399	189
Mean	75.82	74.69	75.24
Median	75.55	75.10	75.80
Standard Deviation	5.00	7.45	7.48
IQR	6.15	10.30	9.70

The first important observation is that the **average age is remarkably similar across all three diagnosis groups**. The difference between the highest and lowest mean is only about 1.13 years.

This is interesting because age is commonly considered an important factor in Alzheimer's disease research. However, in this particular sample, the participants with different diagnoses have very similar average ages.

The second observation is the difference in variability. CN has a standard deviation of 5.00, compared with 7.45 for LMCI and 7.48 for AD.

This suggests that the cognitively normal participants in this sample are more concentrated around their average age, while the LMCI and AD groups contain a wider range of ages.

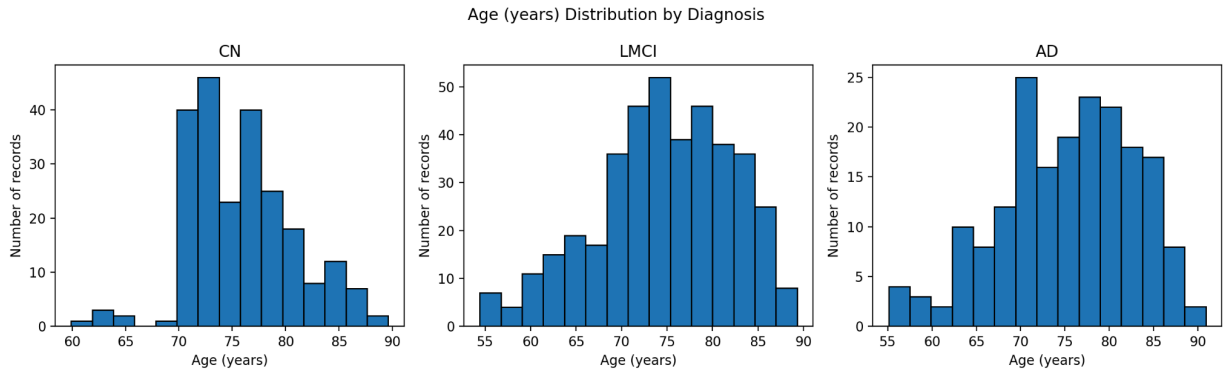
The third observation is the overlap between the distributions. The middle 50% of each group occupies a similar age range, and their medians are nearly identical. This means there is no obvious age threshold that separates CN from LMCI or AD.

Summary:

Age might still contribute to a predictive model, particularly when combined with other variables. However, based on these descriptive statistics, we should not expect age alone to distinguish diagnosis groups particularly well.

An important question for our later modeling is whether age becomes more informative when combined with education, gender, and marital status.

Research takeaway: Age distributions are similar across the three diagnosis groups, although LMCI and AD exhibit greater age variability than CN.



The histograms provide additional information that the boxplot cannot fully communicate: the shape of each group's distribution.

CN: The majority of participants are concentrated around approximately 70–80 years. The distribution is relatively narrow compared with the other groups. There are a few participants at substantially younger and older ages.

LMCI: The distribution extends from approximately 54 to 89 years. Participants are spread across a wider age range, with a large concentration around the mid-to-late 70s.

AD: The distribution also covers a relatively broad age range, extending from approximately 55 to 91 years. Most participants are concentrated around the 70s and early 80s.

One important point is that the histograms have different sample sizes. LMCI has 399 participants, while CN has 228 and AD has 189. Therefore, the taller LMCI bars partly reflect its larger sample size.

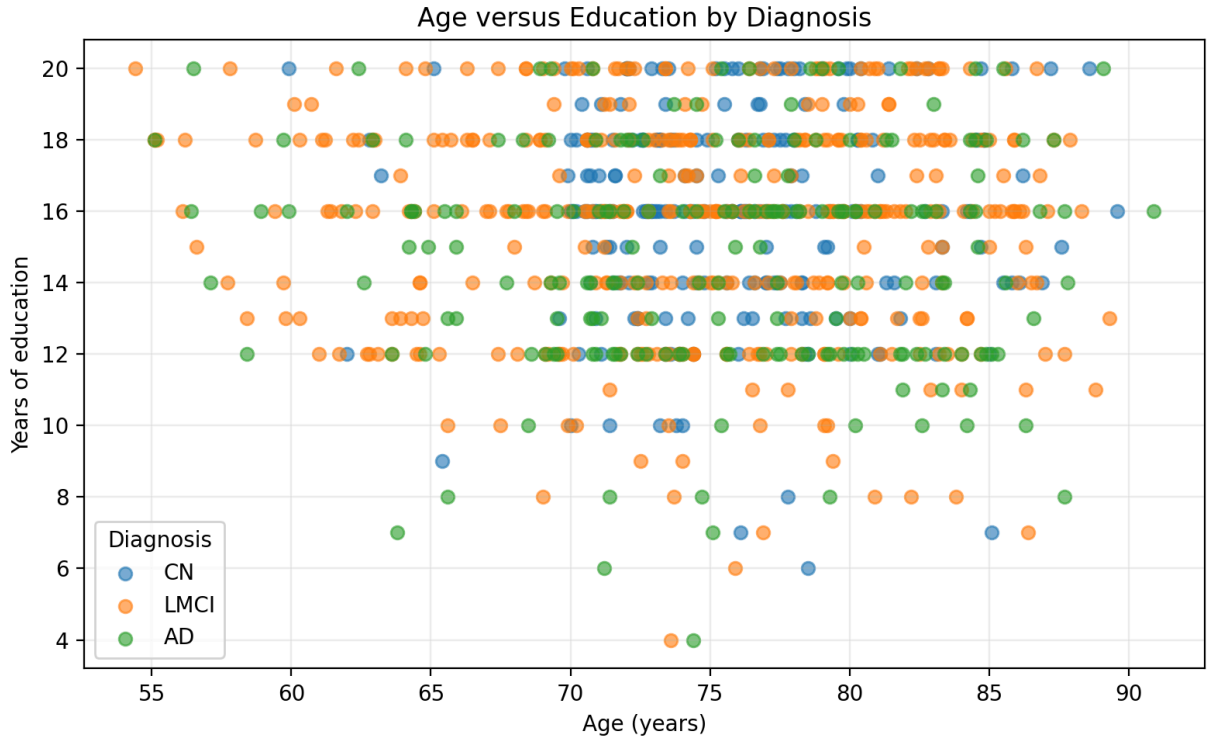
Summary:

The histograms reinforce that diagnosis groups are not separated by distinct age ranges.

For example, participants around age 75 appear in all three diagnosis groups. Consequently, knowing someone's age does not tell us which diagnosis category they belong to.

This also suggests that a decision tree based only on age may struggle to establish useful classification boundaries.

Research takeaway: All three groups have substantial age overlap, despite differences in their age variability and sample sizes.



The most noticeable pattern is the substantial overlap between the three diagnosis groups.

Participants with similar ages and education levels appear across CN, LMCI, and AD. For example, participants around 75 years old with 16 years of education appear in all three categories.

This indicates that there is no visually obvious combination of age and education that consistently separates the diagnosis groups.

However, there is another important observation: education appears to have a different distribution across the groups, particularly for AD.

The scatterplot shows many AD participants around 12–16 years of education, while the CN group has a stronger concentration around 16–18 years. This is consistent with the descriptive statistics we calculated earlier.

Summary:

This graph raises an important distinction:

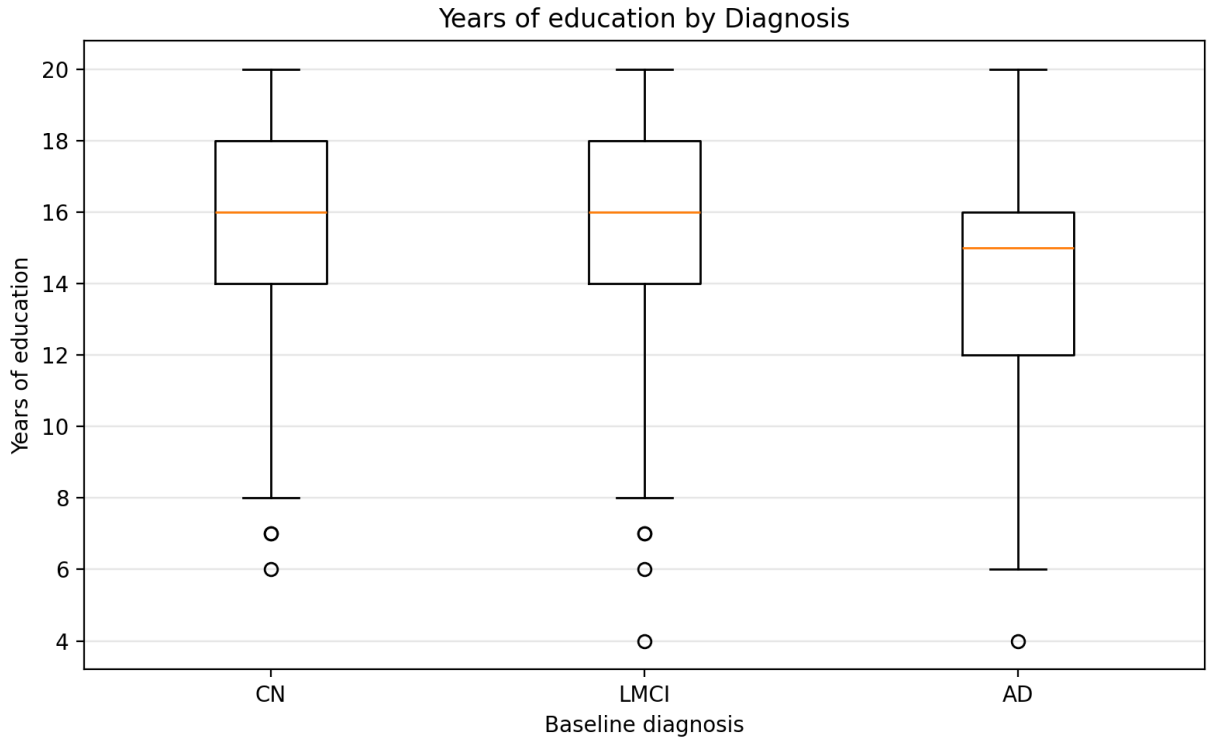
- A variable can have significantly different averages across diagnosis groups.
- That same variable may still have substantial overlap and limited ability to classify individual participants.

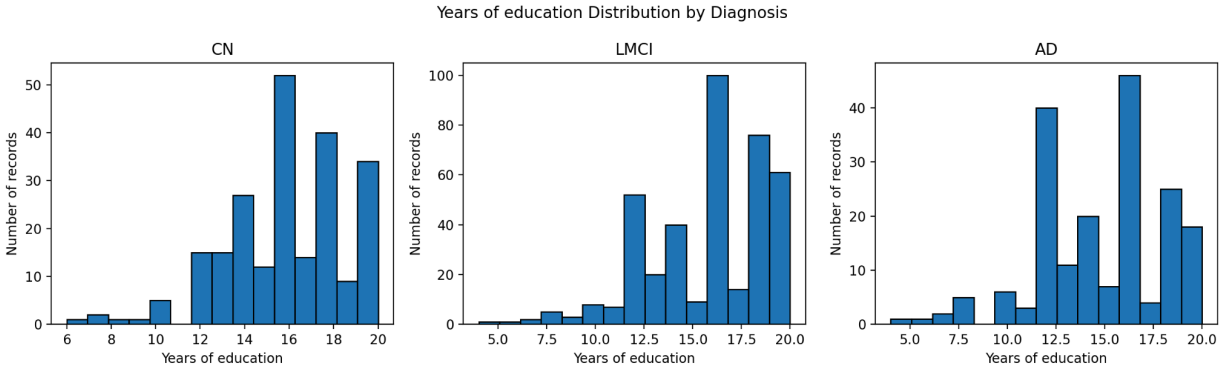
For example, the AD group has lower average education than CN. However, this does not mean that a participant with 16 years of education is unlikely to have AD.

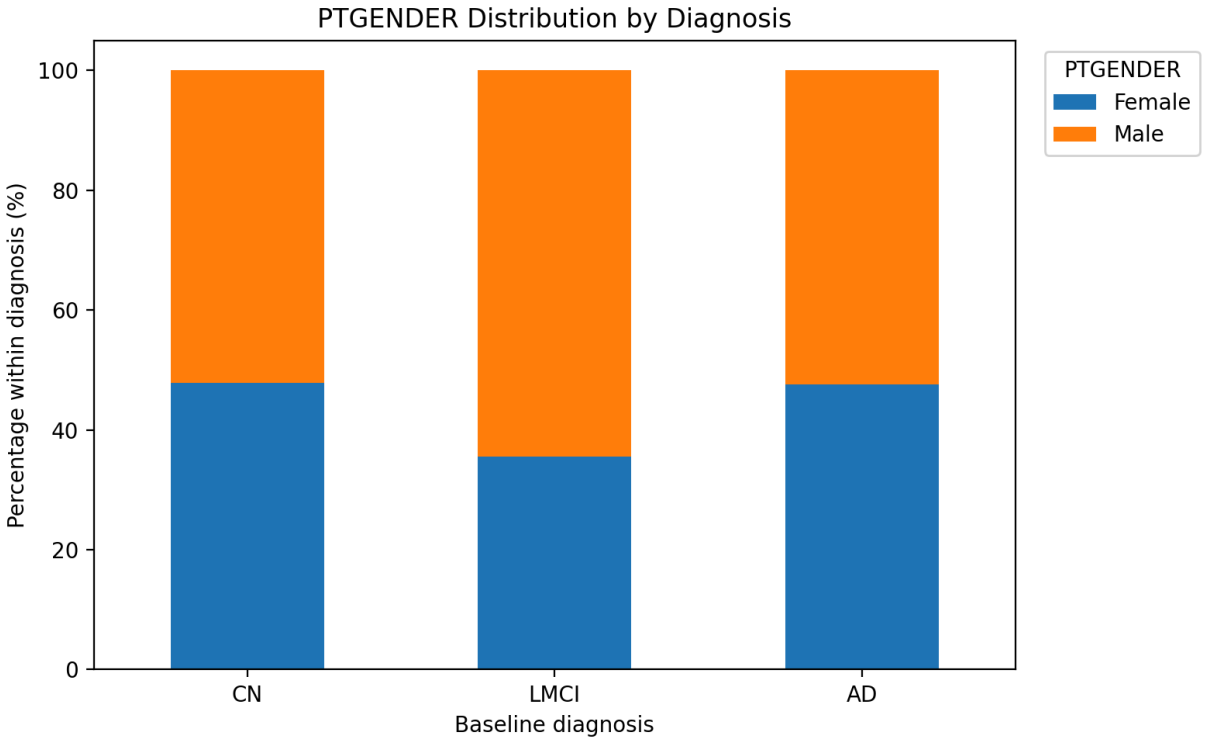
The scatterplot also suggests that simply combining age and education may not produce clear classification boundaries.

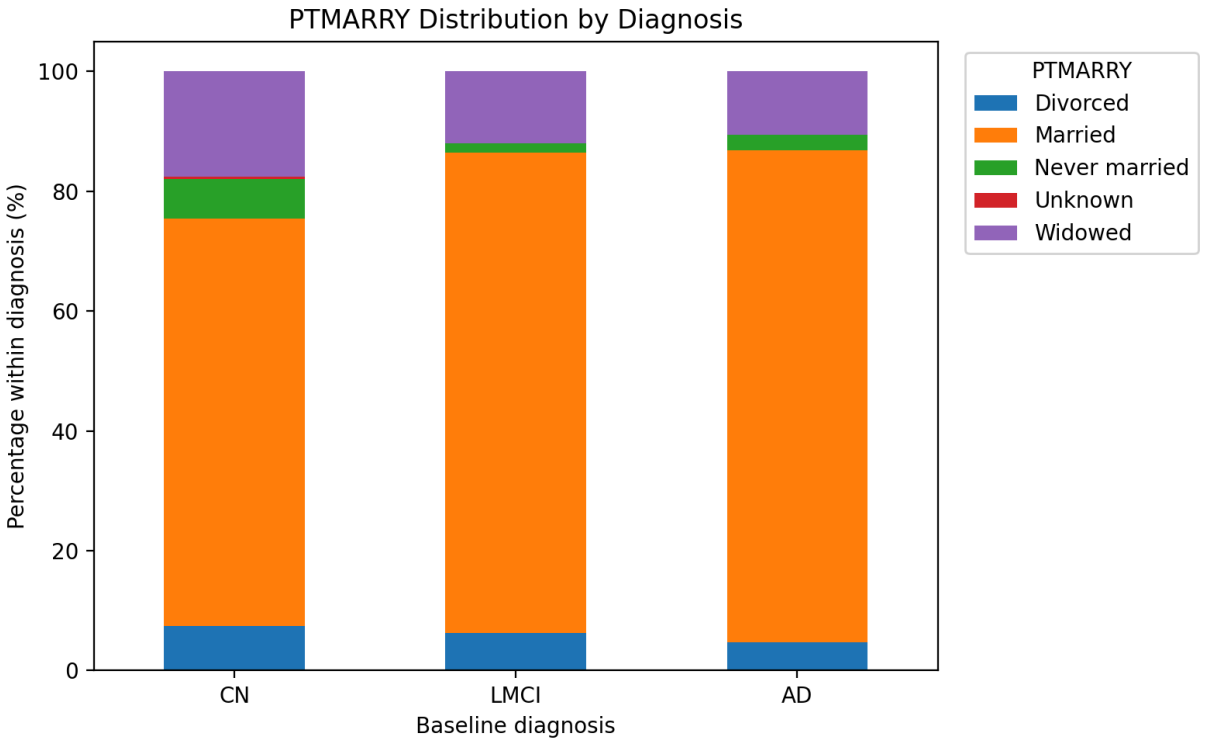
We should therefore investigate whether additional variables improve classification.

Research takeaway: Age and education show substantial overlap across diagnosis categories. Education has noticeable group-level differences, but the combination of these two variables does not visually separate individual participants.


Years of education by Diagnosis
Years of education
20
18
16
14
12
10
8
6
4
CN
LMCI
AD
Baseline diagnosis


Years of education Distribution by Diagnosis
CN
LMCI
AD
Number of records
Years of education


PTGENDER Distribution by Diagnosis
Percentage within diagnosis (%)
0
20
40
60
80
100
CN
LMCI
AD
Baseline diagnosis
PTGENDER
Female
Male


PTMARRY Distribution by Diagnosis
Percentage within diagnosis (%)
0
20
40
60
80
100
CN
LMCI
AD
Baseline diagnosis
PTMARRY
Divorced
Married
Never married
Unknown
Widowed